FUELLING A CLEAN FUTURE

India's transition to piped cooking gas has redefined energy distribution, offering a safer, more efficient alternative to LPG cylinders and polluting fuels

By ANILESH S. MAHAJAN



CHANDRADEEP KUMAR

PIPED GAS DISTRIBUTION
Launched in 1995

Achievement: **Now, 14.2 million households have piped gas connectivity across 23 states. The length of operational pipelines has reached 24,945 km—a 62 per cent decadal growth—with another 10,805 km being constructed**

THE DRUDGERY

The daily struggle of managing liquefied petroleum gas (LPG) cylinders was once a defining feature of urban households in India. From booking to handling, storing and ensuring timely replacements, the process was fraught with inefficiencies. As a net importer of LPG, India's reliance on a sprawling distributor network often revealed gaps in the supply chain. Delays, hoarding, corruption and rationing—typically restricted to one cylinder per 45 days—were widespread, leaving rural and remote areas underserved. These inefficiencies discouraged oil marketing companies (OMCs) from expanding into difficult terrains, forcing many to rely on polluting alternatives like kerosene, coal, firewood and agricultural waste for cooking. The complex subsidy structure, with colour-coded cylinders for residential and commercial use, added to the challenges, fostering systemic corruption and draining the exchequer.

THE EASE OF LIVING

The introduction of piped natural gas (PNG) transformed energy distribution in India, addressing the longstanding issues associated with LPG. The City Gas Distribution (CGD) network delivers gas directly to homes, businesses and industries via an extensive pipeline network, eliminating the need for storage, handling and periodic bookings. It was in 1972 that a domestic consumer got the first PNG connection in Vadodara, years after the discovery of natural gas at Cambay Basin in Gujarat in 1958. But the turning point came in 1995 with the establishment of Mahanagar Gas Ltd, a joint venture between GAIL and British Gas, in Mumbai. Three years later, GAIL joined hands with BPCL and the Delhi government to form Indraprastha Gas Ltd (IGL) for PNG distribution in the national capital. In 2004-05, Petronet LNG established the first gas import terminal at Dahej, followed by Shell's facility at Hazira, both in Gujarat.

Piped cooking gas may be slightly more expensive than subsidised LPG cylinders, but its pay-as-you-consume model alleviates household budgetary concerns

Over the decades, India's natural gas infrastructure expanded significantly. By September 2024, some 24,945 km of cross-country pipelines were operational, with another 10,805 km under construction, linking gas sources to consumption centres nationwide. The PNG network has grown to 14.2 million connections across 23 states, with a dedicated focus on Tier 1 and 2 cities along with their suburbs. This contrasts with 328.3 million LPG connections, over half of which were added in the past decade due to a parallel effort to reduce dependency on unclean fuels in rural and isolated habitations. Now, the government has set an ambitious goal of pushing PNG further into the hinterland and clocking